# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME**: COMPUTER NETWORKS
**COURSE CODE**: CSA0702

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**COURSE OUTCOME COVERED**

***CO4:***
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
*(BL2,BL6)*

*Assessment Tool Weightage: 30%*

19

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**QUESTIONS: 5** **Total Marks:25**

20

Annexure A

## Comparative Analysis

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***Code of Conduct:***

*I K.Yamini (192572191) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

K. Y

# Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Exemplary (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|---|
| Understanding of Concepts | 15% | Demonstrates complete and in-depth understanding of all concepts being compared | Good understanding with minor gaps | Basic understanding; some concepts unclear | Poor understanding of concepts |
| Comparison Depth | 20% | Provides detailed, insightful, and meaningful comparisons across multiple aspects | Adequate comparison with relevant points | Limited comparison; lacks depth | Minimal or incorrect comparison |
| Criteria Selection | 10% | Selects highly relevant and appropriate comparison parameters | Mostly relevant criteria | Some irrelevant or missing criteria | Poor or no criteria selection |
| Analytical Skills | 15% | Strong analysis with logical reasoning and clear justification | Good analysis with some justification | Basic analysis with weak reasoning | No clear analysis |
| Organization & Structure | 10% | Well-organized, clear, and logically structured | Generally organized | Somewhat disorganized | Poorly structured |
| Use of Examples / Evidence | 10% | Uses strong, relevant examples or data to support comparison | Uses some examples | Limited examples | No supporting examples |
| Clarity of Presentation | 10% | Very clear, concise, and easy to understand | Mostly clear | Somewhat unclear | Difficult to understand |
| Conclusion & Insights | 10% | Insightful conclusion with strong justification | Clear conclusion | Basic conclusion | No or weak conclusion |

CO4 - AT3

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1. **Compare Star and Bus Topology in terms of Cost, Reliability and Performance**

**Star topology** connects every device to a central device such as a switch or hub. **Bus topology** connects all devices to a single backbone cable.

**Cost**

- Bus topology is less expensive because it uses a single backbone cable.
- Star topology requires individual cables from every device to the central switch.
- Star also requires a switch or hub, increasing the initial cost.
- Therefore, bus is more economical for a small network.

**Reliability**

- Star topology is more reliable because failure of one cable or computer normally affects only that device.
- In bus topology, failure of the main backbone cable can affect the complete network.
- The central switch is an important point in star topology, so its failure can affect all connected devices.

**Performance**

- Star topology generally provides better performance.
- A switch can control and forward data efficiently to the required device.
- In bus topology, all devices share the same communication medium.
- As more devices are connected, network traffic and collisions can increase.

**Example**

A **school computer laboratory** can use star topology because each computer can be connected individually to a central switch.

**Conclusion**

Bus topology is better for low-cost and simple networks, while star topology is better for performance, reliability and easy management.

## 2. Compare Mesh and Star Topology in terms of Fault Tolerance

**Mesh topology** provides multiple connections between devices, whereas **star topology** uses a central switch or hub.

### Mesh Topology

- Each device has connections to multiple other devices.
- Multiple paths are available for data transmission.
- If one connection fails, another path can be used.
- It provides very high fault tolerance.
- It is suitable for critical communication systems.

### Star Topology

- All devices are connected to a central switch or hub.
- If one computer fails, the other computers can continue working.
- However, if the central switch fails, communication between connected devices can be interrupted.
- Therefore, star has lower fault tolerance than mesh.

### Comparison

| Parameter | Mesh Topology | Star Topology |
|---|---|---|
| Fault tolerance | Very high | Moderate |
| Alternative paths | Multiple | Limited |
| Central device | Not required | Required |
| Reliability | Very high | High |
| Cabling | Very high | Moderate |
| Cost | High | Lower than mesh |
| Maintenance | Difficult | Easy |

**Example**

Mesh topology can be useful in critical networks where continuous communication is important, while star topology is commonly suitable for offices and laboratories.

**Conclusion**

Mesh topology is more fault tolerant than star topology because it provides multiple alternative communication paths.

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### 3. Compare Bus and Mesh Topology in terms of Installation Complexity

**Bus Topology**

Bus topology uses a single backbone cable to which multiple devices are connected.

Advantages during installation:

- Simple network structure.
- Requires less cable.
- Requires fewer connections.
- Easy to install for small networks.
- Low installation cost.

However, finding faults in the backbone can be difficult because one cable is shared by many devices.

**Mesh Topology**

Mesh topology provides connections between multiple devices.

Disadvantages during installation:

- Requires a large number of cables.
- Requires many ports and connections.
- Installation takes more time.
- Network design is more complicated.

- Cost increases as more devices are added.

For **n devices**, a full mesh requires:

**Number of links = n(n − 1) / 2**

For example, if there are 5 devices:

**5(5 − 1) / 2 = 10 links**

So, the number of connections increases rapidly as devices are added.

**Comparison**

| **Parameter** | **Bus** | **Mesh** |
|---|---|---|
| Installation | Easy | Difficult |
| Cabling | Less | More |
| Cost | Low | High |
| Complexity | Low | High |
| Expansion | Limited | Complex |
| Troubleshooting | Backbone faults can be difficult | Many connections make it difficult |

**Conclusion**

Bus topology is simpler, cheaper and faster to install, whereas mesh topology provides greater reliability but requires complex and costly installation.

## 4. Compare Hybrid and Star Topology in terms of Scalability

**Scalability** refers to how easily a network can be expanded when more devices or network segments are added.

### Hybrid Topology

Hybrid topology combines two or more different topologies such as:

- Star
- Bus
- Mesh
- Ring

This allows organizations to design a network according to their requirements.

### Advantages of Hybrid Topology

- Highly flexible.
- Can support large networks.
- Different network sections can use different topologies.
- New sections can be added without completely changing the existing network.
- Suitable for large organizations.

### Star Topology

Star topology can also be expanded by adding more devices to the central switch.

However:

- Expansion depends on available switch ports.
- If all ports are occupied, an additional switch may be required.
- The central device becomes an important part of the network design.

**Comparison**

| Parameter | Hybrid | Star |
|---|---|---|
| Scalability | Very high | Good |
| Flexibility | Very high | Moderate |
| Expansion | Highly flexible | Depends on central device |
| Network size | Large | Small to medium |
| Complexity | High | Low |
| Cost | Higher | Lower |

**Example**

A large university campus may use a hybrid topology by using star topology in individual departments and connecting different departments using another topology.

**Conclusion**

Hybrid topology is more scalable and flexible than star topology because different network structures can be combined according to future requirements.

## 5. Compare Star, Mesh, Bus and Hybrid Topologies in terms of Maintenance

Maintenance involves identifying faults, repairing connections, monitoring devices and keeping the network operational.

**Star Topology**

- Easy to monitor.
- Individual cable failures can be isolated.
- Faulty computers can be identified easily.
- Central switch simplifies network management.
- Therefore, maintenance is relatively easy.

**Mesh Topology**

- Highly reliable but difficult to maintain.
- Contains many connections.
- Troubleshooting requires checking multiple links.
- Repairs can take more time.
- Requires skilled network administrators.

**Bus Topology**

- Simple structure but maintenance can be difficult.
- A backbone cable is shared by all devices.
- A fault in the backbone can affect the entire network.
- Locating the exact fault can be difficult.

**Hybrid Topology**

- Maintenance complexity is moderate to high.
- Different sections may use different topologies.
- Administrators must understand each topology.
- Troubleshooting can vary between network segments.

- Skilled administration may be required.

**Comparison**

| Topology | Maintenance | Main Reason |
|---|---|---|
| **Star** | Easy | Faults are easy to isolate |
| **Mesh** | Difficult | Many connections |
| **Bus** | Difficult | Backbone fault affects network |
| **Hybrid** | Moderate | Multiple topologies are combined |

**Conclusion**

Star topology is the easiest to maintain. Mesh provides high reliability but requires more maintenance effort. Bus is simple but backbone failures are difficult to troubleshoot, while hybrid requires careful management because it combines different topologies.